

Learning from Interrupted Tool Use: Separating Recorded Outcomes from Experience Claims

Working manuscript, incomplete. The methods below are implemented and the baseline observations are reproduced. The complete developmental memory-transfer comparison does not support the proposed instruction's benefit. Independent confirmation and a publication-level contribution are not yet established. This file is not a submission-ready paper.

Abstract

Tool-using agents often retain compressed lessons from past executions. A recorded failure can arise because an agent completed an unsuccessful attempt or because an external execution limit ended an unresolved attempt. Both can legitimately receive zero reward, while supporting different claims about which actions should be avoided in future tasks. We study this distinction through a controlled transfer protocol that keeps the source trajectory, retrieval decision, target task and evaluator fixed. The protocol separately varies whether a curator sees execution-boundary metadata and whether it is instructed to account for that metadata. An initial audit of 40 telecom episodes from two open models finds no instances in which the official task requirements had been fulfilled before an eventual unsuccessful termination. This observation prevents a missing-termination explanation from being used as a general explanation of the observed failures. Offline trajectory cuts nevertheless provide directly observed examples of unresolved prefixes whose recorded continuations later succeed. A complete five-condition transfer study contains 200 episodes over the same 20 development tasks and two checkpoints. Boundary-aware curation, compared with curation receiving the same metadata, produces identical binary outcomes for Qwen3-4B (10/20 in both conditions) and three losses without a gain for Qwen2.5-7B (8/20 versus 11/20). Five protocol exceptions across other curated arms remain unsuccessful in the primary comparison. Qualitative inspection identifies unsupported causal and temporal claims even in boundary-aware lessons. These observations do not support a general benefit from the additional instruction and motivate separating source evidence fidelity from target applicability in subsequent investigations.

1. Introduction

An agent's execution record is an observation under a particular interaction and compute budget. Turning that record into a reusable lesson introduces an additional inferential step. A statement such as “the task was not completed within eight generations” describes an observed outcome. A recommendation such as “avoid this diagnostic procedure because it is ineffective” goes beyond that outcome. The recommendation may be useful when supported by a contradictory tool result, but a deadline alone does not justify it.

This distinction matters because memory is used outside the execution in which it was written. A future task may have different entities, a different initial state, or enough budget to finish a procedure that was previously interrupted. Conversely, extending an execution does not guarantee success: an agent can repeat irrelevant diagnostics, modify the wrong entity, or terminate with a genuinely unresolved problem. A useful evaluation must test the consequences of the resulting memory rather than reward a plausible verbal explanation of the source trajectory.

We investigate this issue using fixed-policy offline cuts followed by cross-task transfer. The source agent does not receive a different budget instruction for each cut. Instead, a completed execution supplies a common history from which observed prefixes are constructed. A curator sees only the selected prefix and permitted execution metadata; the recorded suffix is reserved for analysis. This construction separates a change in observed experience from a change in the policy that generated the experience.

The central empirical question is whether accounting for the origin of termination changes the usefulness of a lesson on a different task. The present experiment is developmental. Its prompt controls do not yet establish a new general memory algorithm, and its small target split cannot support broad claims across agent environments.

2. Problem formulation

Let an execution under a fixed policy produce a history H containing agent generations and environment responses. A generation may include several tool calls. We define H_b as the prefix ending after at most b complete generation groups, including every corresponding tool response. This avoids inventing a partial observation in which an action is visible but its already-produced response is hidden.

Let S_b record whether the prefix ends at the original recorded termination or at an externally imposed cutoff. Let R_b be the supplied completion label: the original recorded reward when its termination is reached, and zero for an externally cut, nonterminal prefix. This assignment follows the pinned evaluator's rule that premature termination receives zero, even when an analysis-only state predicate is already true. The imposed cut is a research intervention, not an independently observed natural termination. We retain the distinction between state satisfaction and benchmark success rather than changing the evaluator or replacing this label after inspecting hidden state.

A curator C produces a memory M from the permitted observation. The target agent executes a different task x with the current official policy and M , and receives the official reward $Y(x, M)$. Our development estimand is the paired change in Y induced by the curation condition while holding the retrieved source episode fixed. It is not the effect of extra source budget, the effect of a new retriever, or a claim about a hidden reasoning process.

The full recorded source continuation can answer a limited descriptive question: did this particular fixed-policy execution later succeed? It cannot prove that every prefix action was beneficial, that every interrupted failure would eventually succeed, or that suppressing a negative lesson improves another task. Consequently, continuation labels are excluded from both memory construction and source selection.

3. Controlled experience transfer

3.1 Source observations and information separation

The source pool is the official telecom train split of tau2-bench. The initial memory pool uses Qwen3-4B-Instruct-2507 trajectories at generation cutoff eight. Cuts at four and sixteen, and a second source model, are retained for later development analyses. Every eligible source record enters the pool without selection by eventual reward. Execution exceptions remain separately counted; they cannot be silently replaced by successful runs.

The visible payload includes the source policy and ticket, observed messages, generation counts, stopping origin and finite-budget reward. Future reward and continuation existence are written to a separate labels directory. The curation program reads only visible inputs and rejects unexpected payload fields. The bank compiler verifies that every source has a generated memory in every curated condition and that the exact prompt matches its registered condition. Missing or failed curation stops compilation.

3.2 Experimental conditions

The no-memory control supplies only the current official policy and ticket. The raw-memory control supplies the selected source ticket, observed actions and responses, boundary metadata and reward, without a generated lesson. The source domain policy is omitted from raw memory because it is already supplied as the current policy.

Three curated conditions share a common instruction to produce a concise lesson containing observations, an actionable recommendation and applicability limits. Outcome-only curation omits stopping metadata. Full-metadata curation adds the observed stopping origin and generation counts. Boundary-aware curation receives exactly the full-metadata evidence, together with an instruction that an external deadline alone does not establish strategy failure and that negative advice requires observed support.

The full-metadata/outcome-only contrast tests additional information. The boundary-aware/full-metadata contrast tests an additional instruction given the same information. All curated conditions use the same checkpoint, greedy decoding and 256-token output ceiling. Actual memory lengths can differ, so equal ceilings do not provide an exact length control. Raw memory can also be substantially longer. Both source curation and target inference costs must therefore accompany any performance comparison. A development quality check found that the initial unconstrained lesson format often exhausted 256 tokens before completing its components. The target pilot was

stopped while only no-memory workers had started. The revised protocol applies a shared three-sentence, at-most-90-word request to every curated arm and refuses to compile a bank containing a token-ceiling output. Original generations and partial baseline records remain archived; this revision was made before any memory-treatment result was collected.

3.3 Retrieval and target execution

One source record is retrieved by TF-IDF cosine similarity between source and target ticket words. Numeric and mixed alphanumeric identifiers are excluded from retrieval tokens. Document frequencies and tie-breaking order are fixed before condition comparison. The retriever does not inspect generated lessons, future rewards, hidden task identifiers or evaluation criteria. Exact ties are recorded because limited ticket diversity can make the selected episode depend heavily on the tie-breaking rule.

The pre-treatment audit finds exact ties for all 20 target tickets, resulting in only four distinct selected source records, used by three, five, six and six targets. Thus the first comparison is conditional on four shared memories. It cannot establish robustness across arbitrary source experiences. Broader claims require outcome-independent variation of the tie selection and must retain dependence induced by reused memories and related target families.

Every memory condition uses the same selected source record for a target ticket. The current official policy and ticket remain verbatim. A shared wrapper marks prior experience as quoted, potentially inapplicable material. The model does not receive the source record ID or condition name. Those values are retained only in the audit trail. The initial target set is the 20-task small split, evaluated separately with Qwen3-4B-Instruct-2507 and Qwen2.5-7B-Instruct.

4. Evaluation protocol and observed baseline

We pin [tau2-bench](#) to commit `b7ea9074c1cba482b30687fecdb5c8425fd6f619` and verify the imported implementation against a source-and-data manifest. We use the official solo telecom transitions and evaluator with 60 message steps and at most five environment errors. Native model chat templates receive the same opening tool-call marker in every condition. Each call permits 512 newly generated tokens and uses greedy decoding. The supplied marker is recorded separately from generated text; it is a formatting baseline, not a proposed research contribution.

The task identities in small, train and test are pairwise disjoint. Hashes of the visible ticket plus complete initial state are also unique within and disjoint across those splits. These checks rule out exact identity overlap, not shared templates or related task families. The test split has not been used for outcome-based selection.

On the completed 20-task small-split baseline, Qwen3 succeeds on 13 tasks and Qwen2.5 succeeds on seven. All 40 episodes have complete traces and no run or protocol exceptions. Qwen3 terminates voluntarily in 19 episodes and reaches the step limit in one. Qwen2.5 terminates voluntarily in 14, reaches the step limit in five and reaches the error limit in one. These descriptive rates are development observations, not treatment improvements or state-of-the-art comparisons.

The fresh no-memory arm under the memory-capable adapter reproduces all 40 prior baseline trajectories. Matching model and task identities, every model input, tool schema, decoded reply and generated token sequence is identical; the official rewards also match. We exclude wall-clock timing and simulation identifiers from this comparison. This checks control reproducibility across the adapter revision, without increasing the number of independent tasks.

Replaying the official state and action requirements at every complete prefix finds that exactly the 20 ultimately successful episodes ever satisfy the requirements. There is no observed success-to-failure reversal and no episode that satisfies the requirements but receives an unsuccessful final benchmark outcome. Thus a missing final submission, observed in a separate synthetic pilot, does not explain the unsuccessful public-benchmark episodes here.

At generation cutoff eight, 14 Qwen3 and 16 Qwen2.5 episodes still have an unobserved recorded continuation. Seven and four of those continuations, respectively, later succeed. These counts establish ambiguity in interpreting some finite-prefix failures. They do not establish that a particular memory method resolves it or improves target reward.

The complete Qwen3 train-source pool contains 74 executions, with 14 official successes and no run or protocol exceptions. Of these, 71 end with agent stop and three at the message-step limit. The same prefix-state replay again finds no satisfied-but-unsuccessful episode and no satisfaction reversal. At source cutoff eight, 69 records are externally interrupted; ten of their recorded continuations later succeed. Four source executions have already ended successfully by that cut. Two of the 69 externally cut prefixes already satisfy the offline state and action checks at generation eight; both eventually terminate successfully. The remaining eight of the ten successful continuations have not yet satisfied those checks at the cutoff. This separates an unfinished termination protocol from an unfinished task state within the constructed prefixes. It does not contradict the absence of satisfied-but-unsuccessful full recorded episodes. The same join across cuts is shown below; its rows reuse the same sources.

Table 1. State satisfaction and observed continuations at three source cutoffs.

Generation cutoff	Externally cut / 74	Already satisfied among cut prefixes	Unsatisfied at cut, recorded continuation succeeds
4	74	0	14
8	69	2	8
16	63	1	6

These are joins to the previously completed strict state replay, not new model executions. Hidden diagnostic labels remain excluded from curation, retrieval and target policy inputs. All 74 sources, including completed unsuccessful attempts, remain eligible for the fixed memory bank. The success rate of this source pool must not be pooled with the small-split target rate or interpreted as a treatment effect.

4.1 Complete developmental transfer results

The registered grid completed all 200 episodes: five conditions, two target checkpoints and twenty target identities. All twenty worker processes exited normally. Five episode-level protocol exceptions occurred within these workers; they remain in the denominator as unsuccessful episodes. We checked the complete condition coverage, source/target identity separation, bank and task hashes, checkpoint and package manifests, and the selected source for every model-task-condition combination before calculating the comparisons.

Table 2. Completed developmental transfer grid with errors retained.

Memory condition	Qwen3 success / 20	Qwen2.5 success / 20	Qwen2.5 protocol exceptions
None	13	7	0
Raw observed experience	11	11	0
Outcome-only lesson	10	7	4
Full-metadata lesson	10	11	1
Boundary-aware lesson	10	8	0

Qwen3 has no protocol exceptions in any condition. In the primary paired boundary-aware/full-metadata comparison, Qwen3 has zero gains and zero losses: the successful task sets are identical. Qwen2.5 has zero gains and three losses, a descriptive decrease of 15 percentage points. The losses concern the Data Saver task and two application-permission tasks. The additional boundary instruction therefore shows no demonstrated benefit in this grid. This does not prove that every treatment effect is zero, nor that boundary information is universally harmful.

Against no memory, raw and full-metadata experience each give Qwen2.5 five gains and one loss. Qwen3 loses two tasks with raw experience and three with each curated arm, without gains. These development observations concern two checkpoints and four retrieved memories. They do not establish general improvement, and the four outcome-only exceptions remain unsuccessful.

Memory transfer on the public development split

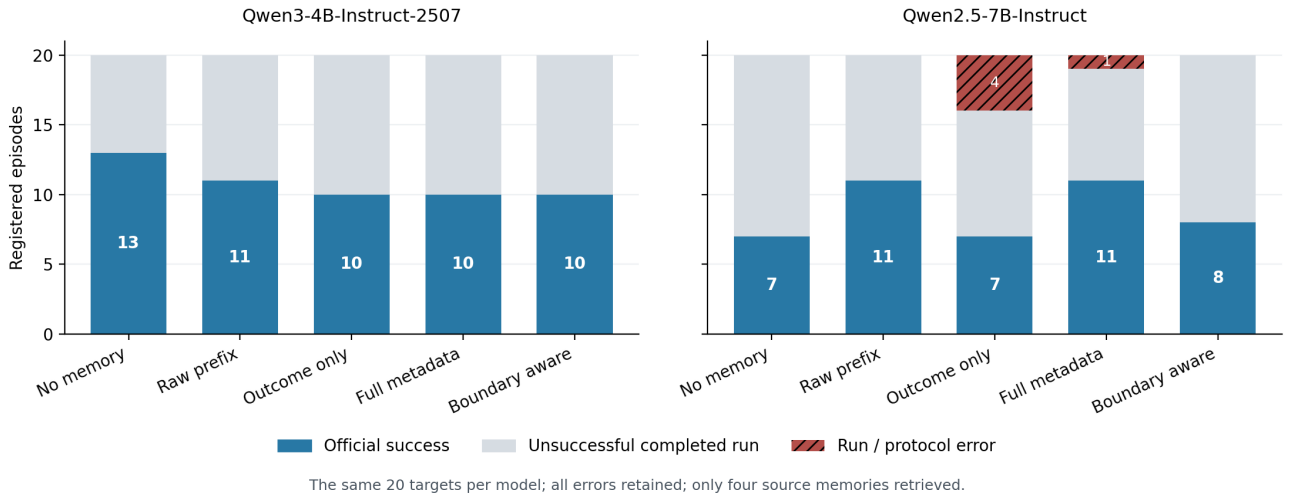


Figure 1. All registered episodes in the developmental transfer grid. Red segments denote protocol exceptions retained as unsuccessful. The bars reuse the same twenty task identities; they are not independent samples of new tasks.

Recorded target generation time sums to approximately 7,160 seconds across the grid; worker wall time sums to 7,870 seconds and overall elapsed batch time is 2,320 seconds. These measure different quantities and are not energy or full-GPU-utilization estimates. Curating the complete 74-record bank costs an additional 248.3, 249.3 and 247.8 generation seconds for outcome-only, full-metadata and boundary-aware lessons, respectively, incurred once per bank and shared across target checkpoints. These values exclude source collection. Lower generation time can reflect early protocol failure and must not be interpreted as efficiency at equal success.

An offline strict replay audits all 155 normally completed memory-treatment trajectories. Exactly their 78 officially successful episodes ever satisfy the task requirements; none loses satisfaction later or satisfies the final requirements while receiving zero reward. For the five protocol exceptions, we reconstruct only the executed history in the final model input, excluding the invalid output. The same reconstruction matches saved simulation prefixes in all 195 normally completed grid episodes. None of the five exception prefixes satisfies the task requirements. Three exceptions emit a status-bar call without an arguments object; two emit a similarly malformed done call. These are protocol failures, but the recorded histories do not support reclassifying them as completed tasks. Diagnostic hidden-state checks occur only after collection and are never exposed to the policy or curator.

4.2 Source evidence and lesson claims

We inspected the three compact lessons for each of the four sources selected by the frozen retriever. These sources were selected before target outcomes, but they are not a random sample of all 74 records. The audit concerns what the observed prefix supports; it does not use the withheld continuation to judge what a curator could have known.

In the MMS source, the agent observes failure, resets the APN, reboots, and observes failure again. All three lessons describe APN misconfiguration as confirmed by the inability to send MMS, although that observation does not identify a cause. The full-metadata lesson goes further and claims the reset resolves the issue, contradicting the final tool observation within its own input. The boundary-aware lesson avoids the explicit success claim but retains the unsupported diagnosis. Thus recognizing an execution boundary does not automatically correct causal overstatement.

A second source checks for a missing SIM and then reseats it. The tool reports successful reseating but the status bar still shows no signal. The boundary-aware lesson describes the SIM as undetected after reseating, even though no post-action SIM-status check appears in the prefix. Lack of signal and lack of SIM detection are distinct state variables. In a third source, a failed speed test becomes an assertion that the remaining issue is beyond agent control, although visible Data Saver and VPN settings have not been ruled out. These errors concern temporal scope and the strength of a claim, rather than missing output punctuation or a malformed memory record.

The fourth source ended successfully after disabling airplane mode and enabling data. Its boundary-aware lesson gives an observation-supported airplane-mode explanation with an explicit applicability condition. A faithful

source summary can therefore coexist with an imperfect retrieval decision: the same generic ticket can describe several different current causes. Source grounding and target applicability should be evaluated separately. The inspection identifies hypotheses for controlled interventions, not causal attribution of a downstream failure to any particular sentence.

4.3 Inspection of the frozen test-selected source lessons

Before obtaining target outcomes for the test extension, we inspected all three compact lessons for each of its fifteen selected source IDs. This is a descriptive review by the same Codex assistant conducting the study, with condition names visible; it is not independent human adjudication. The review records seventeen claim-level examples with exact lesson quotations and references to source tool responses. These examples include supported claims and qualified hypotheses as well as contradictions and unsupported assertions. They are not an exhaustive factuality labeling, so we do not derive an error rate from seventeen examples or forty-five lesson instances. One source pair has identical histories and lessons and supplies no additional independent claim evidence.

The extended inspection identifies an explicit state contradiction: an outcome-only lesson says the device operates on 2G, while every network-type observation in that prefix reports 5G. Its boundary-aware counterpart claims that a data toggle and network-preference change resolved the issue. The only speed test fails between those two actions; no test follows the last action. The claimed post-action resolution is therefore unverified, rather than demonstrably contradicted by a later failed test. Keeping these labels distinct prevents an unknown later outcome from being treated as observed failure.

Another selected source illustrates entity scope. Its ticket identifies a phone number ending in 2002, but the sole account-line lookup returns a different phone number ending in 2001 with account roaming disabled. All three compact lessons state that the user's line is not roaming enabled. The source also observes device roaming disabled, but a device setting does not establish the target account line's state. The recommendation may be worth checking; the asserted target-line observation is unsupported. This is an instance of an existing entity-binding problem, not a new failure class introduced by the boundary intervention.

We also retain counterexamples to a uniformly negative interpretation of cutoff memories. One externally cut source has completion reward zero but ends with an excellent speed-test observation. All three lessons give positive connectivity advice. Another boundary-aware lesson accurately says a speed test still fails after the listed actions and qualifies its recommendation pending further diagnostics. Thus reward zero does not invariably become negative advice, and the boundary-aware condition does not invariably discard failure evidence. These source observations cannot identify which sentence, if any, causes a downstream target outcome.

5. Analysis requirements and limitations

5.1 Frozen test-split extension

After the developmental comparison, we froze a second evaluation on all 40 official telecom test identities. This extension retains the same source bank, curation prompts, target decoding and environment limits. It compares only full-metadata and boundary-aware lessons, with one shared no-memory control per target checkpoint. Neither eventual source reward nor target outcomes enter retrieval. This is an evaluation on previously unused task identities, not an independent environment or a new source-memory sample.

Every test ticket has a tied highest-scoring source set. In addition to the original smallest-record-ID rule, we registered two alternative choices by hash-ranking the remaining tied source IDs with a fixed seed. These choices are deterministic sensitivity settings, not three independent random seeds. They select five sources per rule, reused across target groups of 13, 3, 8, 1 and 15 tasks. All three rules and their paired comparisons will be reported; we do not choose the best-performing rule after collection.

For checkpoint m and registered tie rule r , let $Y(m,r,c,x)$ denote official binary success on target x under memory condition c . The primary contrast is the finite-set mean of $Y(m,record_id,boundary_aware,x)$ minus $Y(m,record_id,full_metadata,x)$ across the 40 targets. The two alternative-rule contrasts are sensitivity analyses. The seven executions per checkpoint and target - one control and six memory treatments - give 560 registered episodes. Protocol exceptions remain unsuccessful. A shared control is never replicated as if three separate no-memory samples had been collected.

Before collecting test outcomes, a literal-content audit found that the 74 source IDs contain only 52 distinct raw memories, 51 outcome-only lessons, 51 full-metadata lessons and 50 boundary-aware lessons. In the group shared by 15 test targets, the two alternative IDs supply identical raw, full-metadata and boundary-aware content. Source IDs and condition names are absent from model input. Thus 60 pairs of registered memory executions have identical memory content across those two rules, rather than providing distinct lesson interventions. We keep both executions and compare complete model inputs, tool definitions, generated token sequences and rewards as reproducibility checks. Timing and source-selection bookkeeping are excluded. Different text hashes elsewhere do not establish semantic diversity or independent evidence.

The primary estimate weights each task equally. An additional descriptive analysis weights the five visible-ticket groups equally and reports each leave-one-group-out estimate. This changes the weighting of the estimand; it does not correct dependence or provide a confidence interval. With only five groups, three deliberately chosen source rules and two fixed checkpoints, we do not interpret these values as population-level uncertainty estimates. The frozen extension is queued; no test outcome or effect is reported here.

5.2 Registered structured-curation extension

A further extension applies the paired boundary instruction within an NK-schema adaptation. The curator retains the prior method's six-field representation and fixed failure vocabularies, but reads telecom prefixes instead of scientific code and logs. It is not a reproduction of the complete Negative Knowledge procedure. Both new arms use the same 1,024-token curation ceiling and the same Qwen3 checkpoint. Visible zero-reward eligibility yields 70 source records. The remaining four successful records are excluded before curation and retrieval, never according to downstream outcomes.

The endpoint comprises all 40 official test tasks, both target checkpoints and both new conditions: 160 registered episodes. These reuse the preceding test task identities and are not an additional independent sample. Frozen ticket-based retrieval selects five sources with reuse 13/3/8/1/15; one task's source identity differs from the original 74-source selection. If a chosen curation is invalid or fails to generate, that source remains selected and an empty experience is supplied under the common wrapper. No alternate retrieval or target exclusion follows from generation quality.

The primary contrast is boundary versus base NK-schema curation within each checkpoint. Its changed pool and larger curation ceiling prevent attributing differences from the earlier generic-curation study to representation alone. Code and task-selection preflights have passed, but new curation and target outcomes are still pending. No accuracy result or benefit is reported here.

5.3 Interpretation limits

The primary development comparison is paired target success for boundary-aware versus full-metadata memories, separately by target checkpoint. The complete grid reports every registered condition, task and failure. Episode audits should distinguish changed entity selection, environment errors, repeated observation, termination and state satisfaction. Syntactic tool validity is not task success.

The 20 development identities remain 20 tasks even when evaluated with several models, cutoffs or conditions. Repeated deterministic executions do not create independent samples. A confirmatory study requires a frozen method and analysis, appropriate family-level dependence handling, uncertainty estimates and an independent evaluation set. Any development tuning must be disclosed.

The current intervention is intentionally simple. Instruction-sensitive gains could reflect wording, lesson length or compression choices. A publishable method claim would require stronger memory baselines and controls addressing these alternatives. The static cross-task reflection control is not an exact reproduction of [Reflexion](#), whose original protocol learns across repeated trials of a task and already includes action-limit-related reflection triggers. No claim of being the first budget-sensitive agent-memory study is warranted.

6. Relation to existing memory methods

[How Memory Management Impacts LLM Agents](#) already studies harmful experience reuse, error propagation and memory selection using subsequent task feedback. Its fixed-memory and deletion comparisons show that source-record quality and downstream usefulness need not coincide. Our observation of unhelpful transfer from a source-supported lesson therefore does not introduce that distinction. The narrower variable here is the treatment of a researcher-imposed execution boundary while holding the source record and retrieval choice fixed. We have not reproduced the prior work's adaptive deletion procedure or established superiority over it.

[Negative Knowledge](#) represents failures with typed layers, scope, evidential degree and recommended actions, and asks downstream agents to justify adoption or rejection. Its schema already allows inconclusive records. Consequently, adding an uncertainty label is not a distinct contribution. A stronger comparison must implement its evidence and adoption structure; our current generic reflection controls do not reproduce that method.

A separate NK-schema adaptation has now been prepared, but has not generated memories or target outcomes. It retains the six-field representation and closed failure vocabularies while changing the input to telecom prefixes. Visible zero-reward eligibility leaves 70 of the 74 source records. This changes the retrieval pool, so its future comparison with the original bank would not isolate representation alone. Its paired boundary-instruction contrast uses identical evidence and a common larger output ceiling. The preparation does not implement the full upstream adoption procedure.

[Grounding Agent Memory](#) lets a post-task curator inspect an environment through read-only probes before updating memory. The present protocol restricts curation to a recorded prefix and its termination metadata. This is a different evidence-access condition, not evidence of superiority. If probing is added in a later experiment, its access to source state and its cost must be explicit; simply adding verification would overlap with that prior method.

[Agent Workflow Memory](#) induces reusable workflows and supports both offline and online use. Our fixed source-bank setting therefore does not itself establish novelty. The narrower variable under investigation is the interpretation of an imposed execution boundary when compressing otherwise matched experience. Whether that variable supports a useful method or a general empirical finding remains to be tested.

[Auditing Self-Evolution in Financial Agents](#) compares memory and workflow systems using matched acquisition trajectories and identifies an execution-interface mismatch in an AWM port. Thus matching source evidence and checking the native action interface are established evaluation concerns. Our controlled curation study must also separate these implementation effects from memory content. We have not reproduced that financial-agent evaluation.

7. Discussion

7.1 Completion labels constrain claims, not recommendations by themselves

The cutoff audit separates three questions: whether the evaluator awards completion, whether the task's state and action checks are already satisfied, and whether the recorded continuation eventually succeeds. These variables cannot be substituted for one another. In particular, zero completion at an imposed cutoff does not identify an ineffective action. Conversely, eventual success does not validate every action in the preceding prefix. A lesson can accurately describe an unsuccessful intermediate test and still recommend a reasonable next step. Our selected-source review includes such cases; it does not support a universal conversion from zero reward to negative advice.

Boundary metadata resolves the origin of the supplied label, but it supplies no missing environment observation. A curator told that execution was cut short still cannot know the unobserved result of a final intervention. This explains the logical limitation of the treatment without claiming to explain its observed target losses causally. The qualitative review finds temporal overstatement in some boundary-aware lessons, but the experiment does not intervene on those individual statements while keeping all other words fixed. It therefore cannot attribute a particular failed target to a particular unsupported sentence.

7.2 Source fidelity and target utility require separate evidence

A lesson may faithfully describe one source and remain inapplicable to a different target state. An unsupported lesson can also coincide with a useful target action. Thus neither target reward nor a source quotation alone is a complete measure of memory quality. Our experiment measures official target reward and provides selected source-level examples separately. It does not combine them into an unvalidated quality score. The 17 reviewed examples are illustrations selected after reading all 45 test-selected lessons, not an exhaustive annotation of atomic claims or a factual-error percentage.

Holding source identity fixed makes the instruction contrast interpretable within the current pipeline, but does not isolate a universal mechanism of boundary reasoning. The instruction may change wording, length, selected facts and recommended actions simultaneously. Greedy curation supplies one lesson per source and condition, not a distribution of possible lessons. The current results consequently concern these concrete memories produced by this curator. A claim about memory algorithms generally would require additional curators, matched content controls and other environments.

7.3 Retrieval dependence limits the effective diversity

Task counts do not describe the full diversity of a memory-transfer study. All development targets share only four selected source records; the test extension uses five under each rule. Literal duplicate lessons further reduce the number of different interventions. Repeating an identical lesson under a different source ID is useful for detecting execution instability, but cannot establish robustness to new experience content. The frozen analyses preserve each task and each rule while reporting this reuse explicitly.

The test extension addresses previously unused task identities and selected source-choice sensitivity. It does not remove dependence between related telecom tasks, provide an independent human review, or compare the complete procedures of the closest memory methods. Those are limits on the scientific claim, even if a future completed comparison were to show positive counts.

8. Provisional conclusion

The completed development experiment provides no evidence of a general advantage from adding a boundary-aware instruction to the same curation evidence. The source audit nevertheless demonstrates why a recorded completion label is insufficient to justify claims about action effectiveness, and why boundary metadata cannot replace missing observations. The principal supported lesson is methodological: report source evidence, termination origin, retrieval reuse and downstream task outcomes separately. The registered test extension is still pending, so this conclusion is limited to the completed development study and the descriptive source audit.

Data, code and use of language models

The code, frozen protocols and analysis programs are maintained on the `codex/agent-paper-research` branch of [Jing-XING/autoresearch](#). `reproduce.md` specifies artifact identities and analysis commands; `claim_to_evidence.md` maps the completed developmental claims to their records. The raw execution archives are retained locally and are not yet a complete public release. The test and structured-curation registrations precede their outcome analysis. Their pending results must not be inferred from development.

A language-model assistant performed literature retrieval, implementation, experiment operation, source-claim inspection, analysis and drafting. The qualitative source review was unblinded and has no independent human adjudication. Numerical development results derive from recorded executions and the pinned official evaluator. Author identities, affiliations, funding and accountability declarations remain to be supplied by the responsible human authors; they have not been invented by the assistant.

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